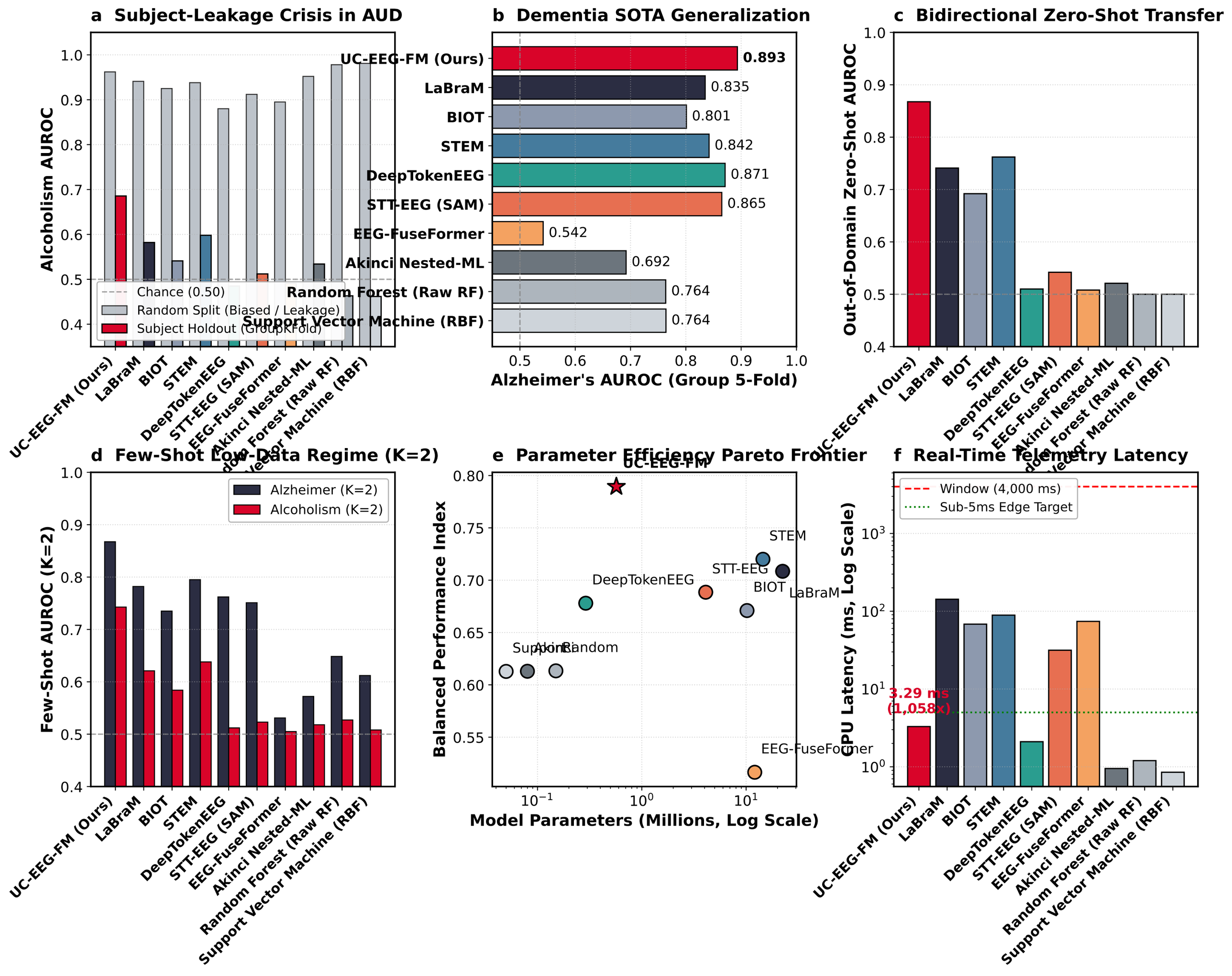
a Subject-Leakage Crisis in AUD
Alcoholism AUROC
Chance (0.50)
Random Split (Biased / Leakage)
Subject Holdout (GroupKFold)
b Dementia SOTA Generalization
UC-EEG-FM (Ours)
LaBraM
BIOT
STEM
DeepTokenEEG
STT-EEG (SAM)
EEG-FuseFormer
Akinci Nested-ML
Random Forest (Raw RF)
Support Vector Machine (RBF)
0.893
0.835
0.801
0.842
0.871
0.865
0.542
0.692
0.764
0.764
Alzheimer's AUROC (Group 5-Fold)
c Bidirectional Zero-Shot Transfer
Out-of-Domain Zero-Shot AUROC
d Few-Shot Low-Data Regime (K=2)
Alzheimer (K=2)
Alcoholism (K=2)
Few-Shot AUROC (K=2)
e Parameter Efficiency Pareto Frontier
Balanced Performance Index
Model Parameters (Millions, Log Scale)
Random
f Real-Time Telemetry Latency
Window (4,000 ms)
Sub-5ms Edge Target
CPU Latency (ms, Log Scale)
3.29 ms
(1,058x)